Stochastic Completion of Undefined Points: A Structural Correspondence Between Gödelian Incompleteness and LLM Hallucination via DIFW

Masaomi Chiba

2025-12-09

Abstract

This paper establishes a rigorous structural correspondence between two forms of computational partiality: (1) the "Gödelian gap" of true but unprovable statements in formal arithmetic, and (2) the "LLM semantic gap" in which a large language model (LLM) fails to produce a grounded completion due to insufficient training support or internal inconsistency. We show that these gaps admit a structural homomorphic mapping based on dependency-graph isomorphisms. We then introduce the Differance Intelligence Framework (DIFW), a stochastic mechanism that fills such undefined points using random semantic proposals evaluated via a coherence-energy functional. This stochastic completion behaves like a probabilistic extension analogous to "adding new axioms" in the Gödelian sense, producing inductive truth approximations that converge to low-energy, high-coherence states. This generalizes Monte Carlo integration techniques into the domain of semantic inference.

1 Preliminaries: Partial Functions and Undefined Points

Let T be a consistent, recursively axiomatized formal system capable of expressing arithmetic. Let M denote a large language model with a learned semantic distribution $\mathcal{D}$ over sequences.

We define:

- The deductive partial function

$$f_{\mathrm{proof}} : \varphi \mapsto \begin{cases} \text{true} & \text{if } T \vdash \varphi, \\ \text{false} & \text{if } T \vdash \neg\varphi, \\ \text{undefined} & \text{otherwise.} \end{cases}$$

The undefined region is the set

$$U_{\mathrm{G}} := \{\varphi \mid f_{\mathrm{proof}}(\varphi) = \text{undefined}\}.$$

- The LLM completion function

  $$f_{\mathrm{LLM}}(s) := \text{the model's next-token completion}$$

  which is undefined (or yields hallucination) when the prompt s lies outside the support of $\mathcal{D}$.

  The undefined region is

  $$U_{\mathrm{L}} := \{s \mid f_{\mathrm{LLM}}(s) \text{ is semantically unsupported}\}.$$

Both partialities reflect a failure of deterministic inference: logical in the former, inductive in the latter.

2 Dependency Graphs and Recursive Obstructions

For a formula φ, let $D(\varphi)$ denote the proof-dependency graph—a directed graph whose nodes are subformulae and whose edges represent recursive or logical dependencies. Gödelian undecidable sentences exhibit non-well-founded structures in $D(\varphi)$.

For an LLM prompt s, let $\mathrm{Dep}(s)$ denote the semantic dependency graph derived from attention traces, token constraints, and contextual referencing. Undefined behavior arises when $\mathrm{Dep}(s)$ exhibits unsupported or self-referential semantic cycles.

3 Structural Correspondence Between Gödelian and LLM Gaps

We now present the central theoretical contribution.

3.1 The Mapping

We construct a mapping

$$\Phi : U_{\mathrm{G}} \longrightarrow U_{\mathrm{L}}$$

such that $\Phi(u)$ is the set of LLM prompts whose dependency structure mirrors the recursive obstruction of u.

Theorem 1 (Structural Correspondence of Undefined Points)**.** *There exists a surjective homomorphic mapping*

$$\Phi : U_{\mathrm{G}} \to U_{\mathrm{L}}$$

preserving the dependency-graph structure responsible for partiality. Explicitly:

$$\Phi(u) := \{\, s \in U_{\mathrm{L}} \mid \mathrm{Dep}(s) \cong D(u) \,\}.$$

Proof Sketch. Gödelian undecidability arises from recursive cycles or unsupported self-reference in $D(u)$. Similarly, LLM undefined points arise when $\mathrm{Dep}(s)$ exhibits structurally analogous cycles or falls outside the learned support of $\mathcal{D}$.

Define $\Phi(u)$ as the class of prompts s whose dependency graphs are isomorphic to $D(u)$. This ensures that Φ preserves the structure of partiality rather than relying on trivial cardinality arguments.

Homomorphicity follows because inference composition in T aligns with semantic resolution composition in M under graph correspondence.

Surjectivity follows since every unsupported structure in $\mathrm{Dep}(s)$ corresponds to some recursive obstruction in T.

Thus Φ maps deductive partiality to inductive partiality via their shared recursive architecture. □

4 Coherence Energy and Stochastic Semantic Completion

We now formalize how DIFW fills undefined points using random semantic proposals evaluated by a coherence-energy functional.

4.1 From Deterministic Failure to Stochastic Filling

The intuitive phenomenon:

$$A \wedge B \to \text{true} \quad \text{(Gödelian failure / deterministic incompleteness)}$$

$$A \wedge \text{garbage} \to \text{hallucination} \quad \text{(LLM failure)}$$

is unified by introducing a *random semantic proposal* X:

$$A \wedge X \to \text{Randomness.}$$

When X is sampled from the DIFW proposal distribution $X \sim \mathcal{P}(s)$, the result is a probability distribution over semantic completions.

4.2 Semantic Distribution Over Undefined Points

For $s \in U_{\mathrm{L}}$, DIFW defines a random variable

$$X_s \sim \mathcal{P}(s)$$

representing candidate completions drawn from a broadened semantic space.

This yields the *semantic distribution*

$$M^{\mathcal{O}}(s) := \{X_s^{(1)}, X_s^{(2)}, \ldots\},$$

analogous to Monte Carlo proposals.

4.3 Coherence Energy Functional

We define a coherence-energy functional

$$E(x) := \alpha \cdot \mathrm{SC}(x) + \beta \cdot \mathrm{CL}(x) + \gamma \cdot \mathrm{OD}(x)$$

where

- $\mathrm{SC}(x)$ = semantic coherence,
- $\mathrm{CL}(x)$ = causal likelihood,
- $\mathrm{OD}(x)$ = observational density (empty-space prior).

The goal is:

$$x^* = \arg \min_{x \in M^{\mathcal{O}}(s)} E(x).$$

4.4 Inductive Truth Approximation

DIFW defines the *inductive completion*:

$$f_{\mathrm{DIFW}}(s) := \mathbb{E}[X_s]_{\text{weighted by } e^{-E(x)}}.$$

This is a probabilistic analogue of extending T with new axioms. The expected value converges to low-energy, high-coherence states.

Theorem 2 (Convergence to Coherence Minimum)**.** *Under mild regularity assumptions on $\mathcal{P}$, the DIFW update*

$$x_{t+1} = x_t - \eta \nabla E(x_t)$$

with random restarts converges in probability to a local minimum of E.

This generalizes Monte Carlo integration: Monte Carlo integrates functions; DIFW "integrates semantic space" to approximate inductive truth.

5 Conclusion

Gödelian incompleteness and LLM hallucination both arise from failures of deterministic inference. Their undefined regions admit a structural homomorphic correspondence based on recursive dependency graphs. DIFW provides a probabilistic mechanism for filling such gaps using a coherence energy functional, generalizing classical Monte Carlo ideas to semantic inference.